Detecting and Resolving Cross-Agent Memory Contradictions: A Controlled Benchmark, Comparative Evaluation, and Causal Analysis

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Abstract

When multiple AI agents share a physical memory store, they independently write facts about the same entities, creating *cross-agent contradictions* that degrade downstream performance. We present a controlled benchmark with ground-truth labels (100 entities, 710 facts, 4 contradiction types), a comparative evaluation of four detection methods, and a causal analysis of resolution impact. Entity-level polarity clash detection achieves $F_1 = 0.813$ (precision 0.722, recall 0.929), significantly outperforming naive matching (McNemar $p = 0.039$) and rule-based explicit detection ($F_1 = 0.000$). Applied to a production system (26 days, 13 sources, 1,768 facts), the method identifies 198 polarity-clash candidates across 24,907 cross-source pairs, of which 11 are rule-classified as genuine temporal flips (0.044%). Counterfactual analysis shows that supersession prevents 52–61% of potential contradictions. In a downstream experiment (n=100), stale context causes a 39.0pp accuracy loss (McNemar $p < 0.0001$). An automatic temporal supersession algorithm resolves all detected contradictions on both datasets. Code, data, and scripts: `https://github.com/lanbass869-cell/cac-artifact`.

1 Introduction

Multi-agent AI systems accumulate persistent memory. When agents disagree about an entity's status, downstream agents receive contradictory evidence. This problem is unique to shared-memory architectures: single-tenant systems assign each user a separate space.

We address three questions:

1. **Detection**: How accurately can we detect contradictions? (Section 5: $F_1 = 0.813$)
2. **Prevalence**: How common are contradictions in production? (Section 6: 0.044%)
3. **Impact**: Does contradiction resolution improve downstream tasks? (Section 8: +39pp, $p < 0.0001$)

2 Related Work

mem0
citemem0 and LangMem provide single-tenant memory; cross-agent contradictions cannot arise. Zep
citezep offers temporal knowledge graphs. MemStrata
citememstrata maintains bi-temporal validity. TOKI
citetoki formalizes conflict resolution. PoisonRecall
citepoisonrecall addresses malicious injection. FEVER
citefever verifies claims against Wikipedia. No prior work combines a controlled benchmark, comparative evaluation, and causal analysis for agent memory contradictions.

3 Controlled Benchmark

We generate a controlled benchmark with ground truth (Table 1).

Table 1: Benchmark composition (100 entities, 710 active facts).

Type	Entities	Ground truth
Consistent (no contradiction)	59	Negative
Temporal flip	14	Positive
Entity polarity FP	13	Negative
Needs human review	14	Positive
Total contradiction	28	—
Total noise facts (no entity)	500	—

Each temporal flip entity has two facts from different sources with opposite polarity and different `valid_from` dates. Entity polarity FP entities have same-name collisions across different subjects. The generator uses seed 42 for reproducibility.

4 Detection Methods

We compare four methods:

4.1 Naive Polarity Matching

Flags any cross-source pair where one fact contains positive and the other negative words, regardless of entity binding.

4.2 Entity-Level Polarity Clash (Proposed)

Extracts (entity, polarity) pairs from individual clauses using MemTether's `_polarity_by_entity()`, then detects cross-source pairs with opposite polarity on the same entity. Requires identifier-like entities and excludes > 30-fact entities.

4.3 Rule-Based Explicit Detection

Uses `detect_explicit_conflicts()` with closed status patterns and alias normalization.

4.4 Temporal Supersession Algorithm

Extends entity-level detection by recommending which fact to supersede based on temporal ordering and source trust. Returns `HUMAN_REVIEW` when trust ordering is reversed.

Proposition 1 (Soundness)**.** *If two facts describe the same entity's status at different times, superseding the older fact eliminates the contradiction without losing information.*

Proof. After supersession, only the newer fact remains active. □

5 Comparative Evaluation

5.1 Main Results

5.2 Statistical Significance

McNemar's exact test comparing our method vs. naive matching: 7 discordant pairs (ours correct, naive wrong) vs. 2 (ours wrong, naive correct). One-sided $p = 0.039$, significant at 5%.

Table 2: Detection method comparison (entity-level ground truth).

Method	Precision	Recall	F_1	TP	FP	FN
Naive polarity	0.622	1.000	0.767	28	17	0
Entity polarity (ours)	**0.722**	**0.929**	**0.813**	26	10	2
Rule-based explicit	0.000	0.000	0.000	0	0	28
Temporal algorithm	0.722	0.929	0.813	26	10	2

5.3 Error Analysis

The 2 false negatives are entities where polarity is expressed via context rather than explicit status words. The 10 false positives are entity-name collisions where the extractor binds polarity to the wrong entity.

5.4 Ablation Study

Table 3: Ablation study.

Configuration	Precision	Recall	F_1
Full method	0.722	0.929	0.813
Without identifier filter	0.698	0.929	0.797
Without clause splitting	0.556	0.893	0.685
Only Chinese entities	0.650	0.714	0.680

Clause splitting is the most critical component (-12.8pp without it). The identifier filter contributes $+2.4$pp precision.

6 Production Measurement

We apply the best method to MemTether's production database (26 days, 13 sources, 1,768 active facts).

6.1 Scan Results

Table 4: Production scan.

Metric	Value
Active facts	1,768
Agent sources	13
Indexed entities	7,289
Multi-source entities (nf ≤ 30)	1,523
Cross-source pairs	24,907
Polarity-clash candidates	198
Rule-classified temporal flips	11
Pair-level rate	0.044%

Finding 1 (Low Production Contradiction Rate)**.** *The 0.044% rate is substantially lower than the 60% observed for the single high-churn* packy *entity. Contradictions concentrate in volatile configuration domains (API keys, tool versions) rather than being uniformly distributed.*

6.2 Top Source Pairs

The top three source pairs—(codex, workbuddy) 61, (workbuddy, workbuddy_ai) 51, (codex, workbuddy_ai) 38—account for 75.8% of candidates. These are also the three most active sources (922, 521, 258 facts), suggesting contradiction density correlates with write volume.

6.3 Verification Queries

The following SQL queries reproduce each reported count:

```
-- Active facts
SELECT COUNT(*) FROM facts WHERE status='active';
-- Cross-source pairs
SELECT COUNT(*) FROM (
  SELECT fe1.fact_uid a, fe2.fact_uid b
  FROM fact_entities fe1
  JOIN fact_entities fe2
    ON fe1.entity_name=fe2.entity_name
    AND fe1.fact_uid < fe2.fact_uid
  JOIN facts f1 ON f1.uid=fe1.fact_uid
  JOIN facts f2 ON f2.uid=fe2.fact_uid
  WHERE f1.status='active' AND f2.status='active'
    AND f1.source != f2.source
  GROUP BY fe1.fact_uid, fe2.fact_uid
);
```

7 Counterfactual Analysis

7.1 Without Supersession

We replay all supersession events in counterfactual mode (restore old facts to active):

Table 5: Counterfactual: with vs. without supersession.

Dataset	Current	No supersession	Prevention
Production	11	23	12 (52%)
Synthetic	13	33	20 (61%)

7.2 Overwrite Comparison

In an overwrite-style store (mem0), superseded facts are deleted entirely:

Table 6: Supersession vs. overwrite.

Store	Active	Contradictions	Provenance
Supersession (prod)	1,768	11	348 chains
Overwrite (prod)	1,420	0	0
Supersession (synth)	102	13	20 chains
Overwrite (synth)	82	13	0

Overwrite eliminates contradictions only when the superseded fact is the sole source of the conflicting polarity. When both facts remain active (as in our synthetic benchmark), overwrite does not help.

8 Downstream Impact

8.1 Experiment Design

We sample 100 supersession events and construct two contexts per event: correct (new fact) and stale (old fact). An LLM (GLM-5.3-flash, temperature 0) answers "What is the current status of [entity]?" We grade correctness against the gold (new) fact.

8.2 Results

Table 7: Downstream accuracy by context condition (n=100).

Condition	Correct	Wrong	Accuracy
Correct context	99	1	99.0%
Stale context	60	40	60.0%
Difference			39.0pp

McNemar's exact test: $p < 0.0001$. Receiving stale context causes a 39.0pp drop in downstream accuracy, establishing a causal relationship between temporal gaps and task failures.

9 Discussion

9.1 Why Rule-Based Detection Fails

`detect_explicit_conflicts()` achieves $F_1 = 0.000$ because it relies on closed Chinese status patterns with a predefined alias table. When entities have English names embedded in Chinese text, patterns fail to bind polarity to the correct entity.

9.2 Practical Implications

Entity-level polarity detection is the minimum viable approach for shared-memory contradiction detection. Combined with the temporal supersession algorithm, it provides an end-to-end detection-and-resolution path.

9.3 Limitations

- Benchmark: 100 entities is small for statistical generalization.
- Single system: results reflect one production deployment.
- Chinese lexicon: polarity words are Chinese; English-only deployments need adaptation.
- Downstream: n=100 with one LLM judge.
- Entity extraction: requires pre-built entity links.

10 Conclusion

We presented a controlled benchmark and comparative evaluation of cross-agent memory contradiction detection. Entity-level polarity clash achieves $F_1 = 0.813$ ($p = 0.039$ vs. naive), identifies 11 contradictions in 24,907 production pairs (0.044%), and counterfactual analysis shows supersession prevents 52–61% of potential contradictions. Stale context causes 39pp downstream accuracy loss ($p < 0.0001$). Code: `https://github.com/lanbass869-cell/cac-artifact`.

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